Inference-time proofreader guidance

Can the released proofreader improve complete contact maps by changing sampling, looking ahead, maintaining several prefixes, repairing errors, or combining larger independent rollout pools?

Frozen generator: exp277 step 266344. Frozen proofreader: exp356 step 18000.
Implementation and GPU validation are in progress; no guided-policy result yet.

First result: aggregation baseline

On existing validation rollouts, sum q remains strongest among five simple aggregators: all-contact R-precision 0.5315; long-range R-precision 0.5046. Frequency gives 0.5167 and 0.4877 respectively. Mean q alone gives 0.4895.

With up to eight rollouts, the pool contains about 75% of native contacts. This separates missing-contact generation from ranking errors. Oracle pool coverage is an upper bound, not the performance of an inference method.

Frozen pilot and execution

32 validation proteins from distinct homology groups span 33–937 residues. Compare top-percentile gates, soft weighting, lookahead, population search and one-pass repair, plus ordinary vLLM pools of 8, 32 and 128 rollouts.

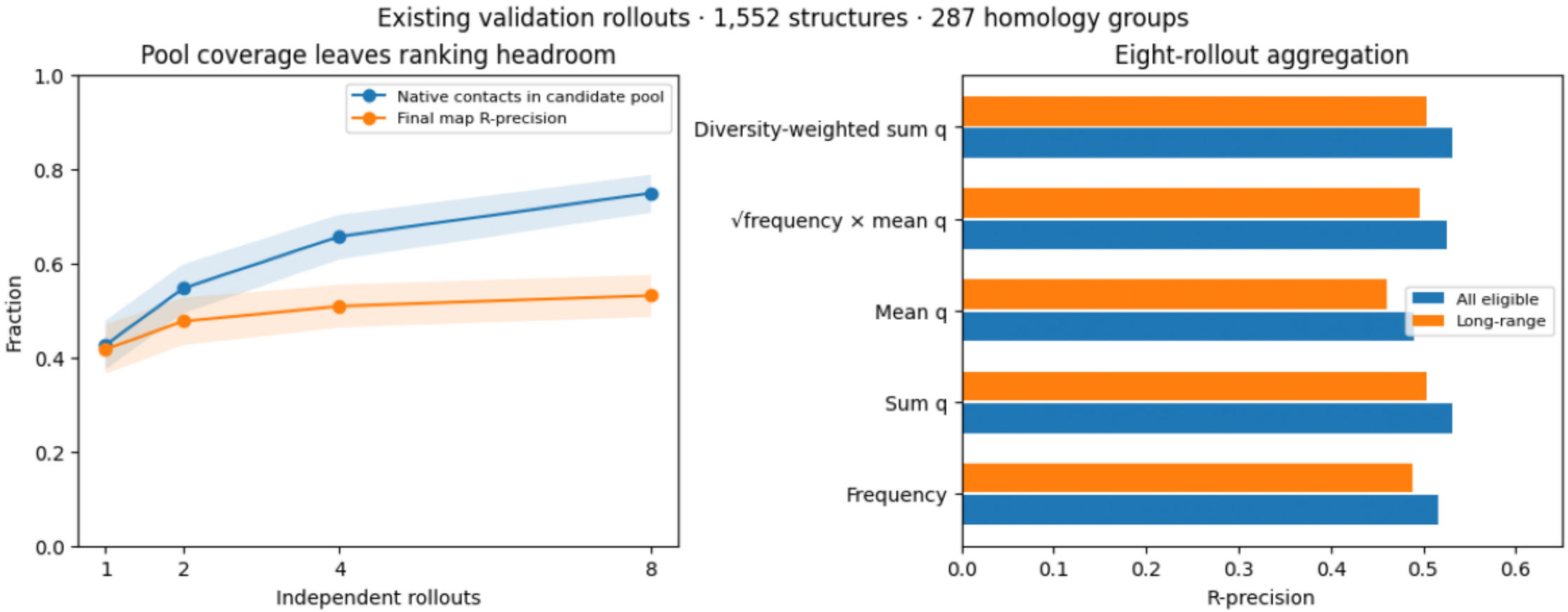
Measure full-universe AP, all/long-range R-precision and complete search cost. Select policies on validation, then lock a reserved-test confirmation plan. Jobs use co-located CoreWeave H100s and resumable per-rollout artifacts.

Numerical validation

All frozen backbone tensors match the generator exactly. CPU cache tests agree with full forward passes. The first H100 smoke caught BF16 incremental/full-pass drift of up to 0.02394 in one contact probability. FP32 cached search passes on three real proteins, with maximum error 0.00000358. Final maps use the released BF16 full-pass reader for comparison.

existing_validation.png

Validation-only reanalysis. Bands resample homology groups (95% intervals). Oracle coverage uses labels only to bound achievable recall. The small number of structures with fewer than eight usable rollouts retains all available samples. Long-range means separation ≥24.




```
$ plot_existing.py
```